Factor-base designs: the m = 3, t = 5 cell on p_2 (run-20261019T1907Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only: nothing here prices a pipeline, reports S or shows a speedup. **Protocol:** PREREGISTRATION.md Amendment 7 (registered in commit `bc31ae4a1` before any T3 row), with Amendments 1–6 for everything it does not change.

Why this run

At m = 3, t = 5 on p_1, A11 completes at D = 5 (F3) and A0 completes at D = 7 on the tower engine (T2): the largest A11-versus-A0 difference measured. §5 asks that a reading be replicated on a second prime before it is described as more than a single-prime result. Amendment 6 admitted the tower engine on p_1 only, so p_2 needs its own admission (T3a) before the A0 cell is measured (T3b).

Run

- **Commands:** `fb_designs_slope admit-tower research/factor_base_designs_20261015/runs --prime p2 --file T1-admission-p2` (T3a), then `fb_designs_slope run --prime p2 --engine tower --arms A0 --m 3 --t-min 5 --t-max 5 --budget 3600 --grace 300 --vmem-kb 15000000 --file 'T3-{arm}-m{m}-p2'` and the same with `--arms A11` on `f4_fp` (no `--engine`).
- **Commits measured:** T3a at `bc31ae4a1b902f2a388b417c6e97aa4d845a5c1e`, T3b at `8e945c6e0257da1ba10dfa18ecd74830b6695e4c`, recorded in each row.
- **Host:** "vm", Intel Xeon @ 2.10 GHz, 4 cores, 15 GB, no swap, a Claude Code cloud container. One system at a time.
- **Window:** 2026-10-19, 19:07Z to 19:36Z (UTC).
- **Limits:** F4 budget 3600 s, hard kill 3900 s, `ulimit -v 15000000`. Seeds as in Amendment 2 with the p_2 prime index. G2 (distinct targets) is off, as in F1.
- **Records:** `runs/T1-admission-p2.jsonl`, `runs/T3-A0-m3-p2.jsonl`, `runs/T3-A11-m3-p2.jsonl`, their `.log` files, and `runs/analysis-tower.txt` (native `analyze --tower`).

T3a: admission on p_2

Admitted. All 22 completed planted A0 systems on p_2 (F1: m = 2, t = 2–9 and m = 3, t = 2–4) complete on the tower engine with the committed `f4_fp` D, without deriving 1, with the planted solution kept. The engine took 0.001 s to 6.6 s per system.

T3b: the cell

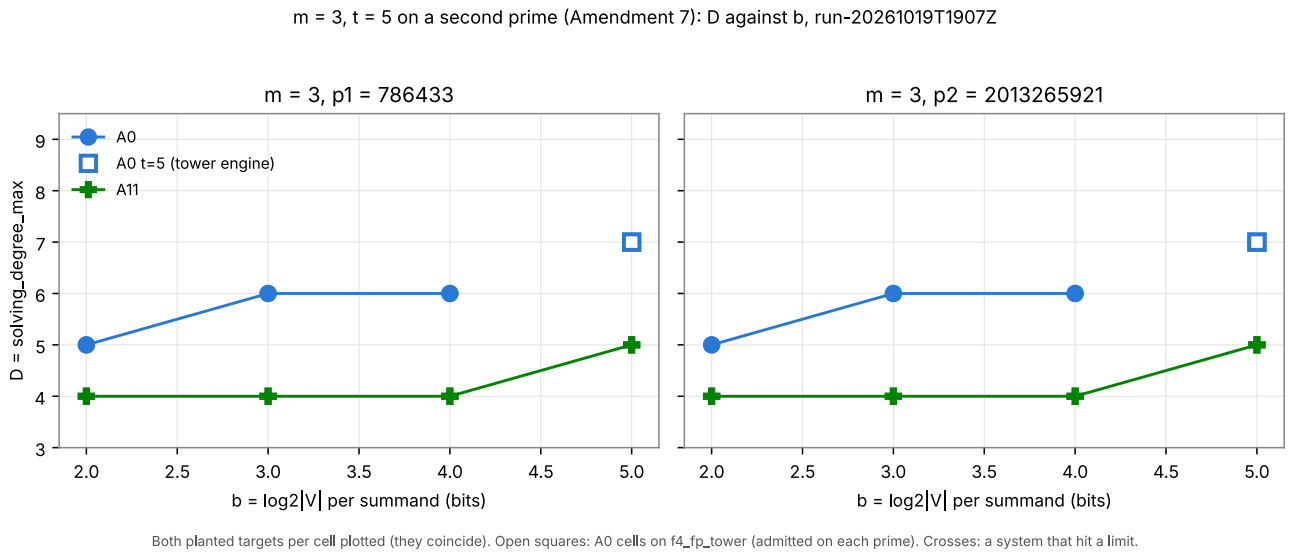


Figure 1: D against b for A0 and A11 at m = 3, on p_1 (left) and p_2 (right). Both targets per cell are plotted (n = 2; they coincide). The open square is A0's t = 5 cell, measured on the tower engine on each prime. Run run-20261019T1907Z; rows from `runs/*.jsonl`, plotted by `plot.py`.

arm (engine)	system	D	degree reached	time (s)	max_cols	status
A0 (tower engine)	target 0	7	8	368.6	37 306	ok
A0 (tower engine)	target 1	7	8	370.4	35 617	ok
A11 (`f4_fp`)	target 0	5	5	486.3	40 486	ok
A11 (`f4_fp`)	target 1	5	5	498.5	40 486	ok

The Amendment 7 rule

"The gap replicates if both cells complete and each arm's cell D on p_2 equals its p_1 cell D (A0: 7, A11: 5)."

arm	p_1 cell D	p_2 cell D	equal?
A0 (tower engine)	7 (T2)	7	yes
A11 (`f4_fp`)	5 (F3)	5	yes

Reading: the gap replicates. Both cells completed, and each arm's D is the same on both primes.

σ on p_2 for m = 3, with the T3 cell as a fourth point (`analysis-tower.txt`):

arm	D over t = 2–5	cells	σ	verdict (§1)
A0	5, 6, 6, 7	4	0.6000	no middle ground
A11	4, 4, 4, 5	4	0.3000	weak

Without the T3 cell, each series has 3 cells: A0 σ = 0.5000 (**no middle ground**) and A11 σ = 0.0000 (**candidate**). On p_2, as on p_1, the fourth cell moves A11′s reading from **candidate** to **weak**.

Predictions scorecard (Amendment 7)

prediction	observed	held?
T3a: admitted	22 of 22 match	yes
both cells complete	4 of 4 systems ok (about 370 s and 490 s)	yes
D = 7 (A0) and D = 5 (A11); the gap replicates	7 and 5; replicates	yes
A0 σ = 0.6000 on 4 cells; A11 σ = 0.3000 on 4	0.6000 and 0.3000	yes

Observations

- The p_1 and p_2 panels are indistinguishable at every cell. This is the same pattern as the 44 earlier systems completed on both primes.
- A11′s `f4_fp` matrix width at the cell is 40 486 on both primes and both targets, to the column. The tower engine's width for A0 is not fixed in that way: 34 706 and 35 617 on p_1, 37 306 and 35 617 on p_2. Only target 1 agrees across primes. That engine's width is not compared with `f4_fp`'s, as the columns count different things. D agrees throughout.
- Replication here means the same two cells on a second prime, with the same construction. It adds a second prime, but not a different kind of instance.

Limits

- As in T2, A0′s D at this cell rests on the tower engine, admitted by agreement on smaller cells (22 on p_2, 26 on p_1). `f4_fp` has no completed system at this cell to compare with.
- Two primes, one cell, two planted targets per cell. Times are wall-clock on one container, and no timing comparison is made between the two engines.
- A11′s two systems at this cell are not guaranteed distinct from each other (G2 is off). Their `x_R` values differ, so they are two systems.

Deviations

None from Amendment 7.